1.11.1 History

Figure 1.51 is the earliest version of a capacitor, called Leyden jar. It is a device that stores static electricity between two electrodes on the inside and outside of a glass jar. It was the initial form of a capacitor. It was invented independently by German scientist Cleric Ewald Georg Von Kleist and by Dutch Scientist Pieter Van Musschenbroek of Leyden city. The invention was named after the city.

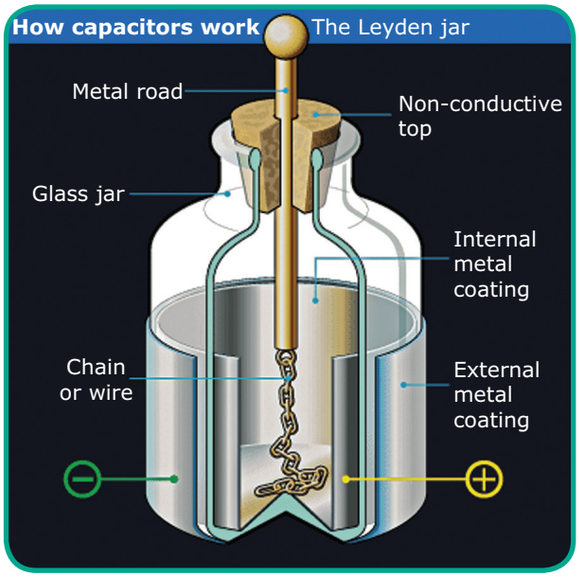


Figure 1.51 Leyden jar

Capacitor was initially known as condenser

1.11.2 Construction

Figure 1.52 shows the parts of the capacitor. In this, an insulator placed between two parallel conducting plates is termed as a dielectric medium. The two parallel plates are termed as electrodes. Based on the insulator used in the capacitors, they are named as paper, ceramic, mica, and polyester capacitors.

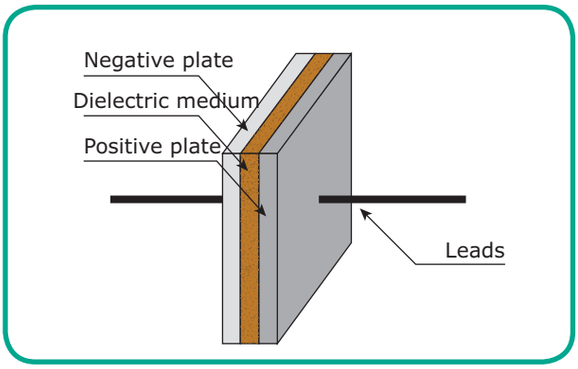


Figure 1.52 Parts of a capacitor

1.11.3 Working of a capacitor

Figure 1.53 shows a capacitor connected in an electrical circuit.

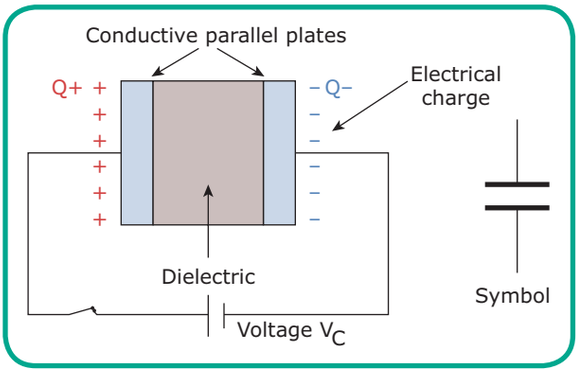


Figure 1.53 Capacitor connected in a circuit.

Charging

When switch *S* is on, current flows through the capacitor. Due to this, the plate *A* of the capacitor gets +ve charge and plate *B* gets –ve charge. Thus, electric field forms across the plates. After complete charging , the current flow will stop. The amount of charge between the plates depends upon the dielectric material and also the distance between the electrodes. The charging limit depends upon the E.M.F. of the battery.

Discharging

If a load is connected to that charged capacitor after removing from the circuit, the electrical energy will discharge. This is known as discharging of capacitor.

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